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Automated and Scalable SEM Image Analysis of Perovskite Solar Cell Materials via a Deep Segmentation Framework

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ABSTRACT

Scanning Electron Microscopy (SEM) is indispensable for characterizing perovskite thin-film microstructure. However, current analysis remains largely manual, limiting throughput and consistency. Here, we present PerovSegNet, an automated deep learning framework for SEM image segmentation that enables accurate identification of lead iodide, perovskite, and defect domains. Built upon an improved YOLOv8x architecture, our model incorporates two novel modules: the Adaptive Shuffle Dilated Convolution Block, which enhances multi-scale feature extraction, and the Separable Adaptive Downsampling module, which preserves fine-scale textures and large-scale structures for robust boundary recognition. PerovSegNet achieves a mean Average Precision of 87.25%, outperforming the baseline by 4.08% while reducing computational cost by 25.22%. Comprehensive evaluations across heterogeneous cross-laboratory datasets and magnification levels demonstrate significant improvements in robustness and scale invariance. Furthermore, digital stress tests reveal enhanced stability under image degradation. Preliminary zero-shot tests on lead-free perovskites, non-perovskite thin films, and cross-sectional imagery indicate strong feature transferability. Beyond segmentation, PerovSegNet provides quantitative grain-level metrics as reliable indicators of microstructural quality, establishing a scalable, data-driven tool for real-time process monitoring and optimization of perovskite fabrication. The source code is available at: <https://github.com/wlyyjj/PerovSegNet/tree/master>.

Introduction

Perovskite solar cells (PSCs) have progressed rapidly since their initial efficiency of 3.9% in 2009, with the state-of-the-art power conversion efficiency (PCE) now exceeding 27% [1]. Their remarkable efficiency gains, coupled with low fabrication cost and solution processability, underscore their potential for scalable and sustainable energy generation [2]. A key factor affecting the performance and long-term stability of PSCs is the microscopic morphology of the perovskite thin film [3, 4]. These films are typically polycrystalline, with crystallinity, grain size and boundary characteristics playing critical roles in charge transport and recombination [5]. Uniform grains and well-defined boundaries enhance carrier mobility, while defects and irregular grain interfaces serve as trap sites that degrade performance and accelerate material degradation under environmental stressors [6]. Therefore, precise characterization and quantification of film morphology, including grain boundaries, defects, and secondary phases like residual lead(II) iodide, which typically arises in iodine-lead-based perovskite systems due to incomplete precursor conversion, are essential for optimizing the fabrication process and improving both efficiency and stability. Scanning electron microscopy (SEM) is widely employed for high-resolution morphological analysis of perovskite films [7–9], which can provide nanoscale insight into surface structure, enabling the identification of key microstructural features. However, manual interpretation of SEM data is labor-intensive, prone to subjective bias, and unsuitable for

high-throughput analysis [10, 11]. As a result, automated and accurate image segmentation techniques have become essential tools for materials characterization [12].

Recent advances in image segmentation methods fall into two main categories: traditional machine learning and deep learning (DL) [13]. Traditional approaches, such as support vector machines [14] and random forests [15], rely heavily on hand-crafted features and thresholding rules. For instance, Yang et al. introduced a progressive grayscale thresholding method to analyze particle morphology in perovskite films, achieving reasonable segmentation results [16]. However, such methods struggle with blurry boundaries and low-contrast regions, often misclassifying lead iodide clusters or merging adjacent grains [17]. In contrast, DL methods, especially convolutional neural networks [18] and YOLO-based models [19], offer significant advantages for analyzing complex microstructures in materials science, which can automatically learn hierarchical representation and perform efficient inference to enable precise and scalable segmentation. For instance, Lilliu et al. combined nano-focused X-ray diffraction with SEM imaging and utilized DL to identify grain boundaries [20]. While this significantly enhanced segmentation accuracy, the method lacked scalability when applied to large datasets involving perovskite and lead iodide. Similarly, Jin et al. employed Electron Backscatter Diffraction alongside SEM imaging to assess grain size [21]. However, the method relied heavily on manual parameter adjustments, compromising its generalizability across diverse perovskite morphologies. In another study, Hidalgo et al. incorporated 3D reconstruction with DL to improve segmentation precision [8]. Despite its accuracy, the approach

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was computationally inefficient when processing complex, multilayered structures, rendering it impractical for real-time or high-throughput applications. YOLOv8, a state-of-the-art real-time detection model, integrates transformer-based modules and an anchor-free design for robust object detection. Despite its strengths, YOLOv8 struggles with the accurate delineation of fine-grained features [22], such as grain boundaries and small defect regions, and its performance remains highly dependent on the availability of large, high-quality annotated datasets [23, 24]. This dependency poses a significant challenge in the context of SEM imaging, where images often suffer from low contrast, background noise, and ambiguous structural boundaries [7, 25, 26], making manual annotation both labor-intensive and error-prone.

To address these challenges, we introduce PerovSegNet, a lightweight yet high-precision segmentation framework tailored for SEM image analysis of perovskite and lead iodide materials. As illustrated in Figure 1, the end-to-end framework encompasses five interconnected stages: (1) SEM image acquisition, (2) assessment of molecular separability via Uniform Manifold Approximation and Projection (UMAP), (3) selection and annotation of regions of interest (ROI), (4) dataset augmentation and model training, and (5) automated segmentation and post-analysis prediction. To overcome the scarcity of annotated SEM data, we constructed an augmented PerovData dataset comprising 10,994 images derived from 46 high-quality, manually labeled samples, with feature separability and morphological diversity verified through UMAP analysis. Built upon the YOLOv8-seg backbone, PerovSegNet integrates two novel modules to enhance segmentation performance: the Adaptive Split-Dilated Convolution with Shuffle Block (ASDCB), which optimizes feature fusion via multi-branch dilated convolutions and attention-based channel recalibration, and the Separable Adaptive Downsampling (SAD) module, designed to capture both fine-scale textures and large-scale structural features. Beyond achieving state-of-the-art segmentation accuracy and computational efficiency, PerovSegNet exhibits strong scale invariance, maintaining robust performance across SEM images of varying magnifications and imaging conditions. Furthermore, it enables automated quantification of grain area and count via post-segmentation analysis, providing a powerful tool for correlating microstructural descriptors with photovoltaic performance and supporting data-driven optimization during device fabrication.

Results

Comparison with Other Models

To evaluate the effectiveness of the proposed PerovSegNet, we conducted comprehensive comparisons against several widely adopted segmentation models, including Mask R-CNN, Cascade Mask R-CNN, and the state-of-the-art Cascade InternImage-XL. All models were trained and tested on the PerovData dataset. Table 1 presents quantitative results in terms of mAP@0.5, mAP@0.5:0.95, parameter count, and

GFLOPs. PerovSegNet achieved an mAP@0.5 of 86.8%, substantially outperforming all baselines. Furthermore, it demonstrated superior computational efficiency with only 87 million parameters and 280 GFLOPs. Compared to models such as Cascade Mask R-CNN and Cascade InternImage-XL, which require significantly more resources, PerovSegNet struck a favorable balance between segmentation accuracy and model complexity.

Ablation Analysis

To assess the individual contributions of the proposed SAD and ASDCB modules, we performed ablation experiments based on the YOLOv8x-seg backbone. The quantitative results are summarized in Table 2, including Precision, Recall, mAP@0.5, mAP@0.5:0.95, parameter count, and GFLOPs. The baseline YOLOv8x-seg achieved an mAP@0.5 of 83.17%. Adding the SAD module improved performance to 86.30%, highlighting the benefits of its dual-path design in preserving fine-scale details. Incorporating ASDCB alone further boosts mAP@0.5 to 86.42%, demonstrating its ability to capture complex multi-scale features. When both modules were combined, the model achieved the highest mAP@0.5 of 87.25% and mAP@0.5:0.95 of 79.96%, while reducing the parameter count and GFLOPs by 24% and 25%, respectively, compared to the baseline. We further analyzed class-specific segmentation accuracy for lead iodide, perovskite and defect regions, and the results are shown in Figure 2. The segmentation accuracy of lead iodide accuracy improved from 84% (baseline) to 88% with both modules, while the segmentation accuracy of perovskite accuracy increased from 82% to 88%. The largest gain was observed for defect segmentation, where accuracy rose from 33% in the baseline model to 64% with both modules. These results confirm that ASDCB strengthens multi-scale feature extraction, whereas SAD preserves fine-scale details critical for defect regions. Their combination enables robust segmentation of both crystalline domains and small irregular features, substantially enhancing performance in SEM image analysis. At the same time, we note that defect detection remains intrinsically challenging. Although the proposed framework significantly improves sensitivity compared to the baseline, missed detections may still occur, particularly for low-contrast or subtle defect regions under cross-system conditions. To further analyze this behavior, the corresponding unnormalized confusion matrices are provided in Figure S1. These raw-count matrices complement the normalized results by revealing the absolute distribution of predictions and errors, and confirm that the observed performance gains are associated with a reduction in background-related misclassifications.

Qualitative Evaluation: Segmentation of Perovskite and Lead Iodide

To further validate model behavior beyond aggregate metrics, we performed a qualitative comparison on representative SEM images that include challenging scenarios, densely packed grains, irregular boundaries, and small defect

Table 1

Comparison of different models on the PerovData dataset. The best results are highlighted in **bold**.

Model	mAP@0.5	mAP@0.5:0.95	Params (M)	GFLOPs (G)
Cascade Mask R-CNN ($\times 101$)	80.4	77.5	122	1493
Mask R-CNN (Swin-T)	82.5	77.9	48	267
Mask R-CNN ($\times 101$)	80.6	78.0	44	447
Cascade InternImage-XL	81.9	79.8	387	1782
PerovSegNet	87.3	80.0	79	265

Table 2

Ablation study of PerovSegNet on the PerovData dataset. The best results are highlighted in **bold**.

Number	SAD	ASDCB	Precision	Recall	mAP@0.5	mAP@0.5:0.95	Params(M)	GFLOPs(G)
1	—	—	97.30	75.31	83.17	76.48	104.67	354.9
2	✓	—	97.38	76.99	86.30	79.31	96.49	340.1
3	—	✓	97.94	78.04	86.42	78.23	87.28	280.1
4	✓	✓	98.11	76.30	87.25	79.96	79.10	265.4

regions (Figure 3). In Figure 3a, Cascade Mask R-CNN ($\times 101$) exhibited segmentation leakage in lead iodide regions, predicted masks spill across grain boundaries, revealing the difficulty of purely convolutional backbones in adhering to fine-scale, irregular contours. Figure 3b showed that Mask R-CNN (Swin-T) recovers some fine structures due to its Transformer features. Nevertheless, despite offering longer-range modeling than standard CNNs, its windowed attention under-captures global context, leading to leakage in larger perovskite domains. Figure 3c (Mask R-CNN, $\times 101$ with convolutional backbones) similarly under-segmented thin lead iodide clusters and yielded high false negatives in low-contrast defect areas. In Figure 3d, Cascade InternImage-XL reduced certain boundary artifacts through multi-stage refinement but still misses small, low-contrast defects, indicating that refinement alone cannot reconcile local detail with broader morphological context. By contrast, Figure 3e (PerovSegNet) delineated sharper boundaries, separates densely packed grains more reliably, and markedly reduces missed lead iodide clusters and defect regions. These gains are consistent with the ablation and class-wise results and stem from the complementary modules: ASDCB strengthens multi-scale representation and boundary localization for small/irregular objects, whereas SAD preserves fine-scale textures and large-scale structures, thereby suppressing leakage and false negatives under complex SEM morphologies.

Robustness Monitoring under Image Degradation

To quantitatively evaluate the practical robustness of PerovSegNet, we performed a systematic digital stress test on the heterogeneous evaluation set (including 5 internal and 22 cross-laboratory images). We simulated three common types of SEM image degradation: (1) Gaussian blurring to mimic defocus, (2) resolution downsampling to mimic loss of spatial detail, and (3) contrast compression to mimic weakened

contrast between lead iodide and perovskite regions. For a comprehensive benchmark, we compared PerovSegNet against three representative models: YOLOv8x-seg, Mask R-CNN (Swin-T), and InternImage-XL (387M parameters). To ensure a fair comparison, all models were retrained on the same perovskite dataset and evaluated under the same degradation settings. As summarized in Table S1 and illustrated in Figure 4, PerovSegNet consistently shows slower performance decay than the comparison models across all three degradation scenarios. In particular, under severe blur, strong downsampling, and reduced contrast, PerovSegNet maintains comparatively higher mAP@0.5 and mAP@0.5:0.95 values, indicating improved robustness under degraded imaging conditions. For example, under Gaussian blur with $\sigma = 5$, PerovSegNet achieves 66.8% mAP@0.5, compared with 64.3% for YOLOv8x-seg, 55.2% for Mask R-CNN (Swin-T), and 51.4% for InternImage-XL. Under 25% resolution scaling, PerovSegNet retains 69.7% mAP@0.5, which remains higher than the corresponding values of the other models. Similarly, under 40% contrast, PerovSegNet still achieves 81.9% mAP@0.5, whereas the other models exhibit larger performance drops. These results suggest that the proposed architecture enables more stable extraction of morphological features under degraded imaging conditions, rather than relying solely on local intensity contrast.

Discussion

PerovSegNet advances SEM image analysis not only by improving segmentation accuracy but also by producing quantitative descriptors that can be directly linked to microstructural characterization. By targeting SEM-specific challenges, including multi-scale grain boundaries, irregular morphologies, and weak-contrast features, the two proposed modules, ASDCB and SAD, improve the reliability of downstream morphology quantification.

Short Title of the Article

System-level demonstration (Figure 5) illustrates the end-to-end workflow in a practical analysis setting. Starting from a raw SEM image, PerovSegNet generates instance masks for perovskite, lead iodide, and defect, together with per-image statistics (areas in μm^2 and instance count). In the example shown, the framework delineates perovskite and lead iodide domains while retaining small, low-contrast defects, consistent with the improvements observed relative to representative CNN baselines. These outputs can be directly used to derive higher-level descriptors, including grain-size distributions and perimeter-to-area ratios (shape complexity), thereby supporting process-oriented analyses such as annealing optimization, precursor adjustment, and morphology-guided quality control. Together, Figures 5 and 7 demonstrate how segmentation outputs can be transformed into predictive indicators for performance-oriented process optimization. By linking microstructural descriptors such as perovskite coverage, lead iodide distribution, and defect density to device-level outcomes, PerovSegNet provides a pathway toward data-driven feedback loops: (i) acquire SEM, (ii) segment and quantify, (iii) consult correlations to anticipate performance impact, and (iv) adjust process parameters to steer morphology toward favorable regimes. While this loop remains prospective, it illustrates the potential for automated SEM analysis to transition from descriptive imaging to actionable process control.

Building upon these quantifiable morphological outputs, we further investigated the model's robustness and scalability, which are critical for practical deployment. The framework's robustness is further validated across scale and contrast (Figure 6, Tables 3 and 4). To assess robustness under heterogeneous imaging conditions, we constructed an evaluation set comprising 27 SEM images, including 5 internal images and 22 cross-laboratory images, and categorized them by both physical scale and contrast. As illustrated in Figure 6, representative challenging regions are shown for qualitative comparison, while the quantitative results are summarized in Table 3 and Table 4. Across four physical scales, PerovSegNet achieves a mean Mask mAP@0.5 of 88.05% and consistently outperforms the YOLOv8x-seg baseline, particularly at low magnification (4 μm), where fine-grained structural details are more compressed. At this scale, the baseline drops to 65.1% mAP@0.5, whereas PerovSegNet retains 76.1%. In addition, the performance variation of PerovSegNet across the four scale categories ($\sigma = 6.98$) is 38.7% lower than that of the baseline ($\sigma = 11.38$), indicating improved cross-scale robustness. The framework also shows improved stability under varying contrast conditions. As summarized in Table 4, PerovSegNet achieves a mean Mask mAP@0.5 of 80.3% across all contrast levels, compared with 75.6% for the baseline. In low-contrast settings, where the SEM contrast between PbI_2 and perovskite regions becomes less pronounced, PerovSegNet maintains stronger performance at both mAP@0.5 and mAP@0.5:0.95. Together, these results indicate that the combined ASDCB and SAD modules improve the stability of morphology extraction across heterogeneous SEM scales and

Table 3

Quantitative performance comparison between the baseline and PerovSegNet across four representative physical scales. The best results are highlighted in **bold**.

Resolution	Baseline (YOLOv8x-seg)		PerovSegNet (Ours)	
	mAP@0.5	mAP@0.5:0.95	mAP@0.5	mAP@0.5:0.95
Very high (200 nm)	93.7	76.4	93.7	84.5
High (500 nm)	88.0	63.8	90.9	67.9
Standard (1 μm)	91.2	84.5	91.5	84.7
Low (4 μm)	65.1	41.5	76.1	48.9
Mean performance	84.5	66.6	88.1	71.5
Std. deviation (σ)	11.4	16.2	7.0	14.7

Table 4

Quantitative performance comparison across heterogeneous contrast levels. The cross-laboratory SEM images were categorized into High, Medium, and Low contrast groups. The best results are highlighted in **bold**.

Contrast level	Baseline (YOLOv8-seg)		PerovSeg (Ours)	
	mAP@0.5	mAP@0.5:0.95	mAP@0.5	mAP@0.5:0.95
High	82.1	70.5	87.2	74.9
Medium	76.3	54.2	82.5	58.7
Low	68.5	46.8	71.3	55.9
Mean performance	75.6	57.2	80.3	63.2

imaging qualities. Beyond the robustness against imaging degradation and physical scale, our framework also exhibits significant resilience when applied to diverse material systems and chemical additives. In practical perovskite systems, passivating agents and functional precursors are frequently introduced to regulate crystallization. While these components do not always form visually distinct domains in top-view SEM, they often influence morphology indirectly. To evaluate the robustness of the framework under such chemical heterogeneity, we benchmarked PerovSegNet on 22 cross-laboratory SEM images collected from peer-reviewed studies. As detailed in Table S2, this set spans a range of material systems and precursor formulations, from standard multi-cation perovskites to mixed-metal and halide-modified compositions. PerovSegNet achieves a Mask mAP@0.5 of 82.1% and a mAP@0.5:0.95 of 56.7% on these unseen samples, indicating that the framework remains responsive to additive- and composition-induced morphological variation. At the same time, explicit identification of an additional material component would require dedicated annotations and task-specific retraining.

Beyond demonstrating technical robustness, it is essential to relate these image-derived structural descriptors to actual device performance metrics. To demonstrate the feasibility of linking image-derived structural descriptors with photovoltaic performance, we curated a meta-dataset comprising 47 matched image-PCE pairs extracted from 20 independent peer-reviewed studies published in 2025 (<https://github.com/Wangyixi-ultra/perovskite-sem>). These studies were selected to ensure the availability of both sufficiently clear top-view SEM images for structural analysis and the corresponding device

PCE reported in the same study. We then applied PerovSegNet model to segment the SEM images and extract quantitative morphological descriptors for correlation analysis. As shown in Figure 7, the correlation heatmap based on Spearman rank correlation (r_s) highlights the relationships between structural descriptors and PCE. Perovskite area shows a strong positive correlation with PCE ($r_s = 0.76$), while perovskite shape complexity (perimeter-to-area ratio) exhibits a slight negative correlation ($r_s = -0.09$). Together, these results indicate that both the extent and spatial uniformity of perovskite coverage are key determinants of device performance, as they jointly regulate light absorption, interfacial continuity, and charge transport pathways. Lead iodide area shows a mild positive correlation with PCE ($r_s = 0.25$), reflecting the complex and dual role of lead iodide in perovskite films. Controlled and discontinuous PbI_2 can promote crystallization and interfacial passivation, whereas excessive accumulation may impede carrier extraction. In contrast, defect-related metrics show consistent negative correlations with PCE, with defect area and defect count correlating at $r_s = -0.22$ and $r_s = -0.31$, respectively, which confirms that defects such as cracks, pinholes, and irregular grain boundaries act as non-radiative recombination centers and degrade device performance.

Furthermore, the model shows promising transferability to emerging lead-free absorbers, such as Sn-based (FASnI_3) [27] and Bi-based ($\text{Cu}_6\text{AgBiI}_{10}$) [28] absorbers. As illustrated in Figures S2 and S3, the model successfully resolves perovskite domains (green) and lead iodide regions (blue) under zero-shot conditions. Notably, the model accurately identified micro-voids as defects (yellow) in the rapidly crystallized FASnI_3 films, while maintaining high specificity in the denser $\text{Cu}_6\text{AgBiI}_{10}$ films. These results indicate that the learned representations capture universal morphological patterns rather than relying solely on high Z-contrast. This highlights the potential of PerovSegNet as a generalized metrology tool for diverse halide perovskite systems.

Finally, we address the limitations and scope of our approach. The reported correlations are observational and should not be interpreted as implying causality. In this study, we focus on three primary morphological descriptors, grain boundaries, residual PbI_2 phases and perovskite grains, because they represent the most fundamental and consistently identifiable features in top-view SEM analysis. However, these descriptors are inherently limited by the 2D projection of surface morphology, and therefore cannot capture 3D characteristics such as grain depth, vertical phase distribution, or grain boundary orientation, which are known to influence carrier recombination dynamics [29]. Accordingly, the extracted features should be regarded as partial indicators of microstructural properties rather than a complete description of device-relevant structural factors. While the present work focuses on top-view SEM images to quantify lateral morphology and surface phase distribution, cross-sectional imaging can provide more direct insight into vertical grain continuity and interface integrity, which are

critical for charge transport [30, 31]. Extending PerovSegNet to cross-sectional SEM data remains challenging due to differences in imaging geometry, contrast mechanisms, and multilayer structures. To explore this direction, we conducted preliminary zero-shot evaluations on cross-sectional images (Figure S4). The results indicate that the model can capture major structural boundaries and grain-like regions despite being trained on top-view data. However, performance is affected by domain differences, and reliable quantitative analysis in this setting would require dedicated training data and task-specific adaptation.

Beyond perovskite thin films, we further explored the applicability of the framework to additional material systems, including chalcogenides (PbS) [32], oxides ($\beta\text{-Ga}_2\text{O}_3$) [33], and Bi-based perovskites ($\text{Cu}_6\text{AgBiI}_{10}$) at a larger scale (Figures S5, S6, and S2). These preliminary results suggest that the framework can transfer across different material systems and imaging conditions, although such transferability remains qualitative and requires further validation with dedicated annotations. Addressing these limitations will require larger and more diverse annotated datasets, as well as the incorporation of uncertainty quantification. Although computational efficiency is improved relative to strong baselines, further optimization and extension toward 3D microstructural analysis will be necessary for integration into real-time in situ or inline characterization workflows.

In summary, this work establishes PerovSegNet as a state-of-the-art segmentation framework for the automated characterization of perovskite thin films. By introducing two modules, ASDCB for enhanced multi-scale feature extraction and SAD for improved preservation of fine structural details, the model achieves state-of-the-art segmentation accuracy while reducing computational cost. Evaluations on cross-laboratory SEM datasets demonstrate that PerovSegNet consistently outperforms strong baselines and large-scale architectures under heterogeneous imaging conditions. Quantitative analysis across a wide range of magnifications (200 nm to 4 μm) shows improved robustness across scales, with a substantial reduction in performance variation compared to the baseline YOLOv8x-seg. In addition, digital stress tests confirm enhanced stability under image degradation, including focus drift and contrast attenuation. Preliminary evaluations on lead-free Sn- and Bi-based perovskites, as well as non-perovskite systems (e.g., PbS and $\beta\text{-Ga}_2\text{O}_3$), suggest that the framework captures transferable morphology-related features beyond the training domain. Exploratory zero-shot testing on cross-sectional SEM images further indicates potential applicability to more complex structural configurations, although dedicated training would be required for reliable quantitative analysis. Beyond segmentation, the framework provides physically interpretable quantitative descriptors, such as phase area fractions and instance counts, which can be used to characterize microstructural quality. These metrics offer a practical basis for linking morphology with device-relevant properties, and may support data-driven process optimization by enabling systematic monitoring of perovskite coverage, lead iodide distribution, and defect density. Despite these

advances, challenges remain in extending the framework to broader material systems and improving inference efficiency for in situ and inline applications. Looking forward, the combination of lightweight design and quantitative analysis capability suggests that PerovSegNet has potential for both laboratory-scale studies and high-throughput screening, contributing to the development of scalable and reliable perovskite photovoltaic technologies.

Methods

Dataset Production

A critical challenge in developing automated analysis methods for thin-film characterization lies in the scarcity of high-quality, expertly annotated perovskite SEM datasets. To address this limitation, we constructed the PerovData dataset, a curated repository for microstructural segmentation. The core of this dataset initially comprised 46 high-resolution internal raw images. The core dataset comprises 46 high-resolution internal SEM images, where the samples were fabricated via a standardized two-step sequential spin-coating process on Glass/FTO/SnO₂/Perovskite multilayer stacks (details in Table S3). Surface characterization was performed using field-emission SEM platforms (Hitachi S-4800 and ZEISS SIGMA HD), with acquisition parameters summarized in Table S4. Image annotation was performed using the open-source tool LabelMe (<https://github.com/wkentaro/labelme>) [34], generating JSON files containing pixel-level segmentation masks and class labels. Representative raw SEM images and their corresponding annotated masks are shown in Figure 8a and 8b. The regions labeled as defects primarily correspond to void-like or pinhole-type structural discontinuities observed in top-view SEM images.

To ensure a fair and leakage-free evaluation, the 46 raw images were first partitioned into training, validation, and internal test sets using an approximate 8:1:1 ratio (36/5/5) at the image-ID level, prior to any augmentation. Consequently, no augmented variants of validation or test images are used during training, ensuring that model performance is evaluated on physically independent SEM images. This internal dataset split was then expanded to 10,994 annotated images using combinatorial augmentation techniques [35]. These augmentations include geometric perturbations, photometric variations, noise injection, blur simulation, intensity transformations, and structural occlusion, mimicking realistic SEM imaging conditions such as focus drift, signal attenuation, and sample tilting. This strategy encourages the model to learn morphology-level features rather than overfitting to repeated local textures. Details of the augmentation pipeline are provided in Table S5. To evaluate robustness under heterogeneous imaging conditions, the internal dataset was complemented with external perovskite evaluation sets. In particular, we incorporated a cross-laboratory image subset consisting of 22 SEM images curated from 20 independent peer-reviewed studies, representing diverse acquisition conditions across laboratories. The overall dataset

organization is summarized in Table S6. For full transparency, comprehensive image-level metadata including material systems, scale bars, image resolution, and contrast levels, are provided in Table S7. The complete dataset is publicly available at <https://github.com/wlyyj/PerovSegNet-Dataset>.

To evaluate the dataset quality and the separability of perovskite, lead iodide and defect regions, we employed UMAP, a nonlinear dimensionality reduction technique that preserves both local and global structure in high-dimensional feature space. By projecting the feature space into two dimensions, UMAP [36] enables intuitive visualization and assessment of class separability. As shown in Figure 8c, distinct clusters emerge for the three classes: perovskite, lead iodide, and defects, demonstrating strong intra-class cohesion and clear inter-class separation. This supports the reliability of the annotated labels and justifies their suitability for supervised DL segmentation.

We further analyzed the geometric and spatial characteristics of annotated instances. Figure 9a shows a histogram of instance counts across categories: deep blue denotes lead iodide, while light blue denotes perovskite. Figure 9b illustrates the width-height distribution of annotated instances using bounding boxes (minimum enclosing rectangles of segmentation masks). By aligning the centers of all boxes to a common origin, we obtain a normalized shape-overlay representation that highlights the diversity of instance sizes and aspect ratios, where darker regions indicate higher overlap frequency. Figure 9c illustrates the spatial distribution of instance centers over the normalized image plane (1.0×1.0), with color intensity indicating local density. Figure 9d shows the equivalent diameter distributions of the identified phases, indicating that perovskite and PbI₂ grains predominantly lie in the submicron regime (peaking around $0.2 \mu\text{m}$), while defect regions are concentrated at smaller nanoscale dimensions (peaking around $0.05 \mu\text{m}$).

Model & Architecture Enhancement

The YOLO series of algorithms represents a class of real-time object detection frameworks renowned for their high inference speed and strong detection performance. Among them, YOLOv8 introduces a refined architecture with the C2f module as its backbone, enabling improved feature extraction. However, this default backbone exhibits limitations when applied to high-resolution SEM images, particularly in capturing fine-grained morphological details and managing complex background noise frequently observed in perovskite and lead iodide films. To address these challenges, we propose PerovSegNet, a customized segmentation architecture built upon YOLOv8x-seg and enhanced with two key modules: the ASDCB and the SAD. The ASDCB module replaces the original C2f component, introducing multiple convolutional branches with adaptive receptive fields to enhance feature fusion. This design significantly enhances the network's ability to discriminate grain boundaries, small particles, and defect regions. In parallel, the SAD module

substitutes the traditional pooling-based downsampling operations, which utilizes a dual-path architecture that integrates depthwise separable convolutions with adaptive pooling mechanisms. This approach effectively mitigates common issues in traditional downsampling methods, such as aliasing and limited sensitivity to small-scale textures.

The overall architecture of PerovSegNet is depicted in Figure 10. It follows a typical detection and segmentation pipeline comprising three major components: Backbone, Neck, and Head. The ASDCB-enhanced Backbone is responsible for hierarchical feature extraction; the Neck performs multi-scale feature fusion to integrate semantic and spatial information; and the Head produces high-resolution segmentation masks with improved delineation of perovskite and lead iodide regions. The structural design and working mechanism of the ASDCB and SAD components are described in detail in the following paragraphs.

In the Backbone, we embed the SAD module at each downsampling position, replacing conventional pooling operations. The SAD module employs a dual-path strategy combining depthwise separable convolutions and adaptive pooling mechanisms. The adaptive design allows the module to dynamically capture fine-scale textures and large-scale structures, which are crucial for retaining the integrity of global features and particle boundaries in SEM imagery. Alongside SAD, the ASDCB module is also repeatedly integrated into the Backbone after each convolutional block. ASDCB fuses dilated convolutions with Squeeze-and-Excitation (SE) operations to capture multi-scale contextual features while enhancing local morphological sensitivity. This dual-focus on fine and contextual information improves the model's ability to discriminate between small particles, grain boundaries, and extended defect regions in heterogeneous materials.

In the Neck, the ASDCB module continues to play a critical role in multi-scale feature aggregation. Unlike traditional designs that apply fixed convolutional blocks in the Neck, our incorporation of multi-branch convolutions enables dynamic receptive field adjustment, enhancing the Neck's ability to propagate high-resolution cues from the backbone while integrating deeper semantic content. Following each feature concatenation and upsampling operation, ASDCB is applied to enrich semantic information and maintain spatial consistency across different resolution levels. This design ensures that both shallow and deep features are adaptively refined before fusion, mitigating the loss of structural detail typically associated with naive upsampling.

In the Head stage, the fused features are passed through the prediction layers, which include the Spatial Pyramid Feature Fusion (SPFF) module and the final segmentation head. SPFF captures hierarchical spatial relationships and improves scale-aware detection. The final 1×1 convolutional layer outputs segmentation maps with class labels, bounding box coordinates, and confidence scores. This architecture enables precise localization and categorization of perovskite grains, lead iodide phases, and defects across a wide range of sizes and shapes.

The architecture of the proposed ASDCB is illustrated in Figure 11. ASDCB integrates SE attention with dilated and depthwise separable convolutions, and further employs group convolution and channel shuffle strategies to enhance both multi-scale feature aggregation and computational efficiency. Given an input feature map X , ASDCB processes it through four parallel branches to extract multi-scale and channel-adaptive features.

The first branch applies standard convolution to extract fundamental spatial features:

$$X_1 = \text{Conv}(X) \quad (1)$$

The second branch employs dilated convolution to enlarge the receptive field and capture broader contextual information:

$$X_2 = \text{DilatedConv}(X, d) \quad (2)$$

where d represents the dilation rate, allowing the model to capture larger receptive field without increasing parameter count.

The third branch introduces a channel-wise attention mechanism using the SE block to adaptively adjust the weights of each channel, enhancing the network's selectivity for important features. This is implemented via global average pooling (GAP), followed by two fully connected layers:

$$S = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \text{GAP}(X))) \quad (3)$$

$$X_3 = S \cdot X \quad (4)$$

Here, σ is the sigmoid activation, and the parameters W_1 and W_2 are the weights of the two fully connected layers. S serves as the learned channel-wise attention vector.

The outputs from all four branches are aggregated:

$$X_{\text{sum}} = X_1 + X_2 + X_3 + X_4 \quad (5)$$

A channel shuffle operation is then applied to improve cross-branch feature interaction:

$$X' = \text{channel_shuffle}(X_{\text{sum}}, g) \quad (6)$$

where g denotes the number of groups. The fused features X' are passed through a Convolutional Feedforward Network (ConvFFN), which consists of two pointwise convolutions and DropPath regularization, with DropPath improving generalization by randomly omitting entire convolutional paths during training:

$$X_{\text{out}} = \text{ConvFFN}(X') \quad (7)$$

The ASDCB module is specifically tailored to address challenges in SEM image segmentation of perovskite and lead iodide films, which often exhibit fine-grained structures, multi-scale morphologies, and complex backgrounds. Conventional convolutional blocks face two major limitations in this context: limited receptive field size and high computational redundancy. ASDCB mitigates these challenges through the following innovations:

- **Multi-branch convolution for multi-scale feature extraction:** The use of standard, dilated, depthwise, and attention-enhanced convolutions allows ASDCB to capture both

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localized and contextual features critical for distinguishing small grains and large structural regions.

- **Channel shuffle for enhanced feature fusion:** By redistributing channels across branches, the model gains stronger interactions between heterogeneous features, resulting in more expressive representations.
- **Computational efficiency via lightweight design:** Depthwise separable convolutions and DropPath reduce computational burden, enabling efficient segmentation on high-resolution SEM data under limited resource constraints.

Complementary to the ASDCB module, the SAD module is further proposed to optimize feature downsampling and fine-detail preservation in the segmentation network.

The architecture of the SAD module is shown in Figure 12. It adopts a dual path design, combined with depthwise separable convolution and adaptive pooling, to jointly capture fine scale (high-frequency) textures and large-scale (low-frequency) structures, where fine-scale details correspond to features such as grain boundaries, edges, and textures, while large-scale structures represent wider morphology and overall film uniformity [37]. Depthwise separable convolutions process each channel independently, making them particularly effective at preserving subtle intensity differences between neighboring pixels, for example, those observed at crystal boundaries or particle edges, thus retaining critical microstructural information. On this basis, the left branch uses stride $s = 2$ for downsampling, focusing on capturing global features or coarse-grained structural information, such as large-scale shape and overall uniformity. The right branch uses stride $s = 1$ to maintain the spatial resolution of the input feature map, and enhances local feature extraction capability through adaptive max pooling and adaptive average pooling. Adaptive max pooling highlights significant local structural features, while adaptive average pooling helps aggregate smoother contextual information. Together, these two pathways enable the SAD module to integrate information across different spatial scales, thereby improving segmentation accuracy for complex perovskite and lead iodide morphologies.

Given an input feature map $X \in \mathbb{R}^{C \times H \times W}$, where C is the number of channels and H, W are the spatial dimensions, the SAD module first applies two depthwise separable convolutions with different strides: one with stride $s = 2$ for downsampling, and another with stride $s = 1$ to retain high-resolution details. These convolutions can be expressed as:

$$X_1 = \text{DepthwidthConv}_1(X) \quad \text{with} \quad s = 2 \quad (8)$$

$$X_2 = \text{DepthwidthConv}_2(X) \quad \text{with} \quad s = 1 \quad (9)$$

Subsequently, the output feature map is further processed by adaptive max pooling and adaptive average pooling, which adjust pooling regions based on the input to capture both fine and coarse features:

$$X_{\max} = \text{AdaptiveMaxPool}(X_2) \quad (10)$$

$$X_{\text{avg}} = \text{AdaptiveAvgPool}(X_2) \quad (11)$$

Table 5

Hardware and software setup used for training.

Name	Configuration / Version
Operating system	CentOS Linux 7.6.1810(Core)
CPU	AMD EPYC 7543 (32-core)
GPU	NVIDIA A100 80GB PCIe
Memory	1.0TB
Graphics memory	80 GB
IDE	Visual Studio Code 1.94
Framework	PyTorch 2.2.2
CUDA	CUDA 12.0
cuDNN	cuDNN 8.9.0
Python Version	Python 3.8.19

Table 6

Training Hyperparameters for YOLOv8 and PerovSegNet.

Parameter	Value
Initial learning rate	0.01
Epochs	300
Batch size	4
Momentum	0.937
Weight decay	0.0005
Warm-up epochs	3.0
Warm-up momentum	0.8
Warm-up bias learning rate	0.1
Input image size	640 × 640
Number of classes	3
Optimizer	AdamW

The results from both pooling operations are then concatenated to form a unified feature map:

$$X_{\text{concatenated}} = \text{Concat}(X_1, X_{\max}, X_{\text{avg}}) \quad (12)$$

The key innovation of the SAD module lies in combining depthwise separable convolutions with adaptive pooling for downsampling, instead of relying solely on static pooling. Traditional pooling methods are computationally efficient but often discard fine details and boundary precision, which are essential in SEM image segmentation. By contrast, the SAD module balances local and global representations, retaining subtle boundaries while preserving large-scale morphology. This adaptive strategy is particularly effective for capturing the intricate features of perovskite and lead iodide microstructures, thereby enhancing segmentation accuracy.

Training Protocol and Evaluation Metrics

All experiments were conducted on a CentOS Linux 7.6.1810 (Core) system, with hardware and software configurations summarized in Table 5. Model development was implemented using PyTorch 2.2.2 with CUDA 12.0 on an NVIDIA A100 GPU (80 GB memory). All input SEM images were resized to 640 × 640 pixels to match the YOLOv8 input requirements. Due to adaptive resizing and padding, the effective input resolution preserves the original aspect ratio, with one dimension scaled to 640 pixels.

To ensure fair and stable comparisons across different model families, two distinct training configurations were adopted according to their respective frameworks. For YOLOv8x-seg and the proposed PerovSegNet (Ultralytics framework) (<https://github.com/ultralytics/ultralytics>), models were trained for 300 epochs with a batch size of 4 using the AdamW optimizer. The initial learning rate was set to 0.01 with a weight decay of 0.0005. A warm-up strategy, including momentum and bias learning rate warm-up, was applied during the early training stage to stabilize convergence. For baseline models implemented in MMDetection (<https://github.com/open-mmlab/mmdetection>) (including Mask R-CNN, Mask R-CNN (Swin-T), and InternImage-XL), a unified optimization configuration was used. All models were trained for 120 epochs with a batch size of 16 using the Adam optimizer, an initial learning rate of 0.001, and a weight decay of 0.0001. The learning rate was scheduled using a StepLR policy with a decay factor of 0.1 every 10 epochs, and gradient clipping (threshold = 5) was applied to ensure stable optimization. Detailed hyperparameter settings are provided in Table 6 and Table S8.

To avoid potential bias during model comparison, hyperparameter tuning was strictly confined to the training and validation sets, while the internal test set remained fully unseen. Model selection criteria were kept consistent across all methods, and framework-specific optimization strategies were applied as summarized in Table S9.

The performance of the proposed PerovSegNet model is quantitatively assessed using six commonly used metrics: Precision, Recall, mean Average Precision at IoU=0.5 (mAP@0.5) [38], mean Average Precision at IoU thresholds from 0.5 to 0.95 (mAP@0.5:0.95), number of trainable parameters, and computational complexity in Giga Floating Point Operations (GFLOPs). These metrics collectively evaluate the detection accuracy, segmentation quality, and computational efficiency of the model. Precision quantifies the proportion of correctly predicted positive instances out of all predicted positive instances, while Recall calculates the proportion of true positive instances out of all actual positive instances, offering insights into the model's ability to identify and classify relevant objects. The mAP@0.5 metric measures the average precision across all classes at an IoU threshold of 0.5, providing a comprehensive assessment of detection performance. The mAP@0.5:0.95 metric extends this by evaluating the average precision across IoU thresholds from 0.5 to 0.95, offering a more stringent measure of detection quality that takes into account the model's ability to match predicted and ground truth objects at varying degrees of overlap. The number of Parameters reflects the overall model complexity, and GFLOPs indicates the computational cost during inference. These metrics enable a thorough evaluation of the model's segmentation capabilities on the SEM dataset of perovskite and lead iodide films, considering both detection fidelity and computational feasibility.

Declaration Statements

Data Availability

The datasets generated and/or analyzed during the study are available on GitHub (<https://github.com/wlyyj/PerovSegNet-Dataset>). Source data are provided with this paper.

Code Availability

The algorithm is written in Python 3.8.0. The corresponding source codes and scripts are available through GitHub at (<https://github.com/wlyyj/PerovSegNet/tree/master>).

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Author Contributions

X.C. conceived, supervised, and funded the project, and provided overall guidance on research design, methodology, and manuscript preparation. L.W. developed the PerovSegNet model and performed the main experiments. T.X.H. constructed and preprocessed the dataset. J.G.P., H.Z., and Y.Q.Z. contributed to technical validation, provided critical feedback, and supported the structural analysis. X.C. and L.W. wrote the manuscript. All authors discussed the results and approved the final manuscript.

Competing Interests

The authors declare no competing financial or non-financial interests.

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Figure Legends

Figure 1: Overview of the PerovSegNet-based segmentation workflow. The pipeline begins with SEM image acquisition, followed by the evaluation of structural separability across molecular species using UMAP. A ROI is selected and annotated to construct a training dataset, which is then augmented and used to train PerovSegNet. The trained model performs automated segmentation and predicts pixel-wise classifications for perovskite, lead iodide and defect.

Figure 2: Normalized confusion matrices from ablation study. a) YOLOv8x-seg baseline, b) YOLOv8x-seg + ASDCB, c) YOLOv8x-seg + SAD, and d) YOLOv8x-seg + ASDCB + SAD (PerovSegNet). The results highlight improved class-wise segmentation accuracy, especially for defect regions.

Figure 3: Qualitative comparison of segmentation on representative SEM image. (a) Cascade Mask R-CNN ($\times 101$), (b) Mask R-CNN (Swin-T), (c) Mask R-CNN ($\times 101$), (d) Cascade InternImage-XL, and (e) PerovSegNet (ours). Cascade/Mask variants exhibit leakage in lead iodide regions (a,c) and missed small defects (d), while Swin-T shows leakage in larger perovskite areas due to limited global context (b). PerovSegNet achieves cleaner boundaries and

fewer missed detections, particularly for small lead iodide clusters and defect regions.

Figure 4: Integrated quantitative robustness and performance decay analysis. (a) Blurring resilience simulating focus drift; (b) resolution stability under downsampling; and (c) contrast robustness simulating secondary electron signal loss. Solid lines represent the mAP@0.5, while dashed lines denote the high-precision mAP@0.5:0.95. PerovSegNet (red tracks) consistently exhibits the shallowest decay slopes across all scenarios.

Figure 5: Automated segmentation and quantification platform. Top-left: original SEM image. Top-right: segmentation map (perovskite in blue, lead iodide in red, defects in yellow). Middle-left: automated statistics reporting lead iodide. Middle-right: automated statistics reporting perovskite. Bottom: analysis panel with runtime settings and logs. The outputs provide per-image metrics for process feedback and quality control.

Figure 6: PerovSegNet segmentation results across SEM images with progressively increasing scale: (a) 200 nm, (b) 500 nm, (c) 1 μ m, and (d) 2 μ m. The red and blue outlines denote segmented lead iodide and perovskite regions, while the white numeric labels indicate the confidence scores of the detected instances. The examples illustrate the ability of PerovSegNet to maintain consistent segmentation behavior across different SEM scales.

Figure 7: Correlation heatmap between image-derived microstructural descriptors and device PCE. Features include the area, count and shape complexity of lead iodide, perovskite and defect. Cells show Spearman rank correlation coefficients, where red indicates positive correlation and blue indicates negative correlation.

Figure 8: Dataset construction and separability evaluation. a) A representative SEM image from PerovData dataset, captured at 20,000 $\times$ magnification, reveals the granular morphology of perovskite films. b) Annotated segmentation map showing individual regions of perovskite (green), lead iodide (red), and defects (blue). c) UMAP projection of the dataset highlighting molecular separability, where clusters correspond to perovskite (purple), lead iodide (dark green), and defects (light green).

Figure 9: Comprehensive statistical characterization and geometric analysis of the PerovData dataset. a) Instance frequency across categories: lead iodide, perovskite, and defect. b) Heatmap showing the size and number of instances across SEM image regions. c) Heatmap of the position distribution of the center point of instances relative to the entire image. The color map indicates the local density of annotated instances, with warmer colors corresponding to regions of higher concentration. d) Statistical distribution of microstructural equivalent diameters for Perovskite, lead iodide, and defect.

Figure 10: Overall architecture of PerovSegNet. The network consists of three stages: Backbone for hierarchical feature extraction, Neck for multi-scale feature fusion, and Head for segmentation map generation. PerovSegNet tailored for SEM image segmentation of perovskite and lead iodide,

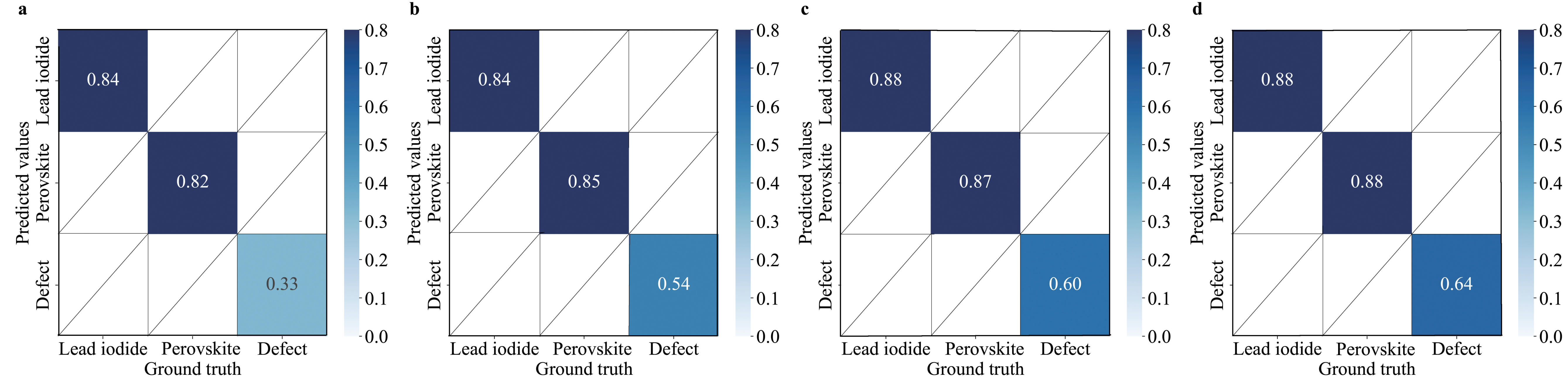
extends the YOLOv8x-seg framework by incorporating two customized modules: ASDCB and SAD, which enhance sensitivity to fine-grained morphology and robustness against complex backgrounds.

Figure 11: Architecture of the ASDCB module. The module is composed of four parallel branches: Standard Convolution, Dilated Convolution, SE layer, and Depthwise Separable Convolution. Each branch is specifically designed to extract complementary features across different scales. The outputs from these branches are fused through a Channel Shuffle operation, which is followed by a ConvFFN for feature integration and transformation. This design enables ASDCB module to enhance the feature representation for complex SEM images of perovskite and lead iodide materials.

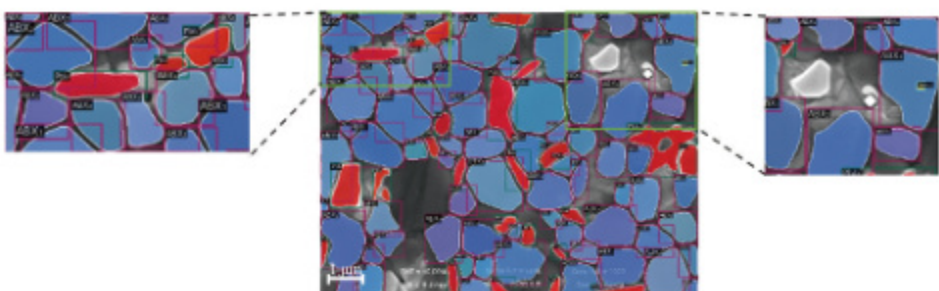
Figure 12: Architecture of SAD module. The module consists of two parallel branches using depthwise separable convolutions with different strides. One branch captures global morphology, while the other is processed through adaptive max pooling and average pooling to preserve high-resolution details. The outputs are concatenated to form a unified feature representation, enabling effective integration of fine details and large-scale structures in SEM image.

Supplementary Information

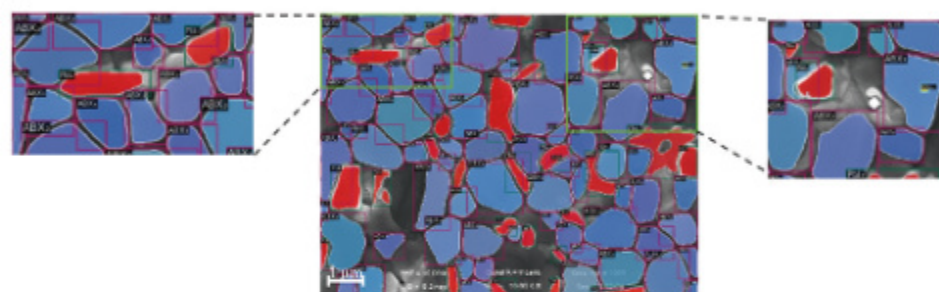
Supplementary data associated with this article can be found in the online version, including: robustness evaluation under non-ideal SEM imaging conditions and dataset chemical heterogeneity (Supplementary Note 1), internal experimental and imaging specifications (Supplementary Note 2), dataset composition and metadata (Supplementary Note 3), model implementation details and hyperparameter configurations (Supplementary Note 4), and additional statistical analyses, cross-system validation, and generalization results on diverse thin-film systems (Supplementary Notes 5–7).



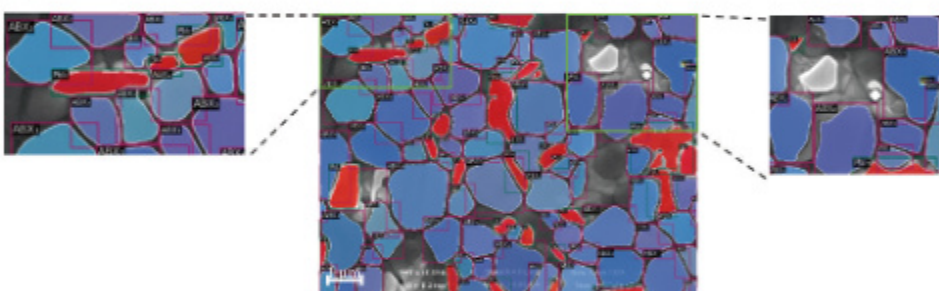
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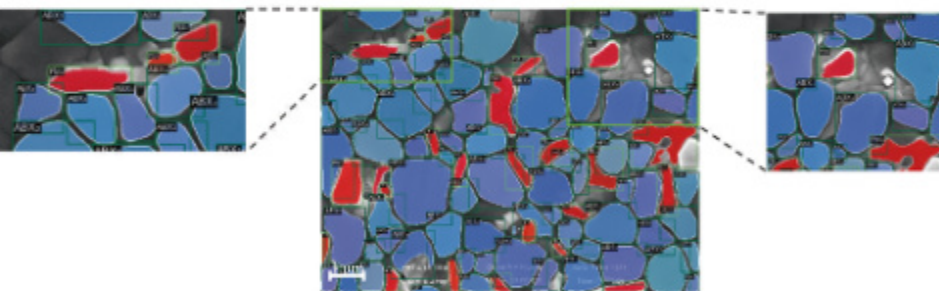
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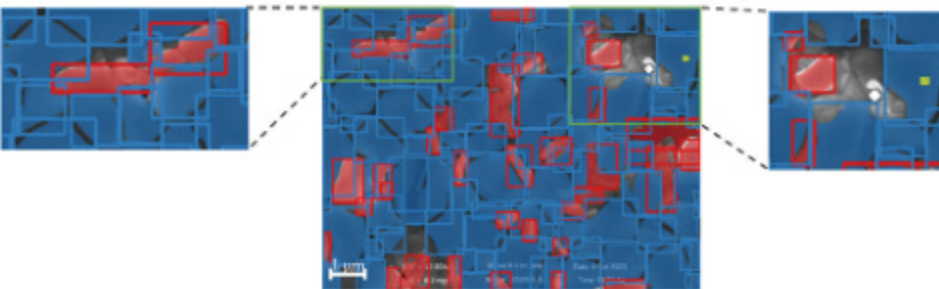
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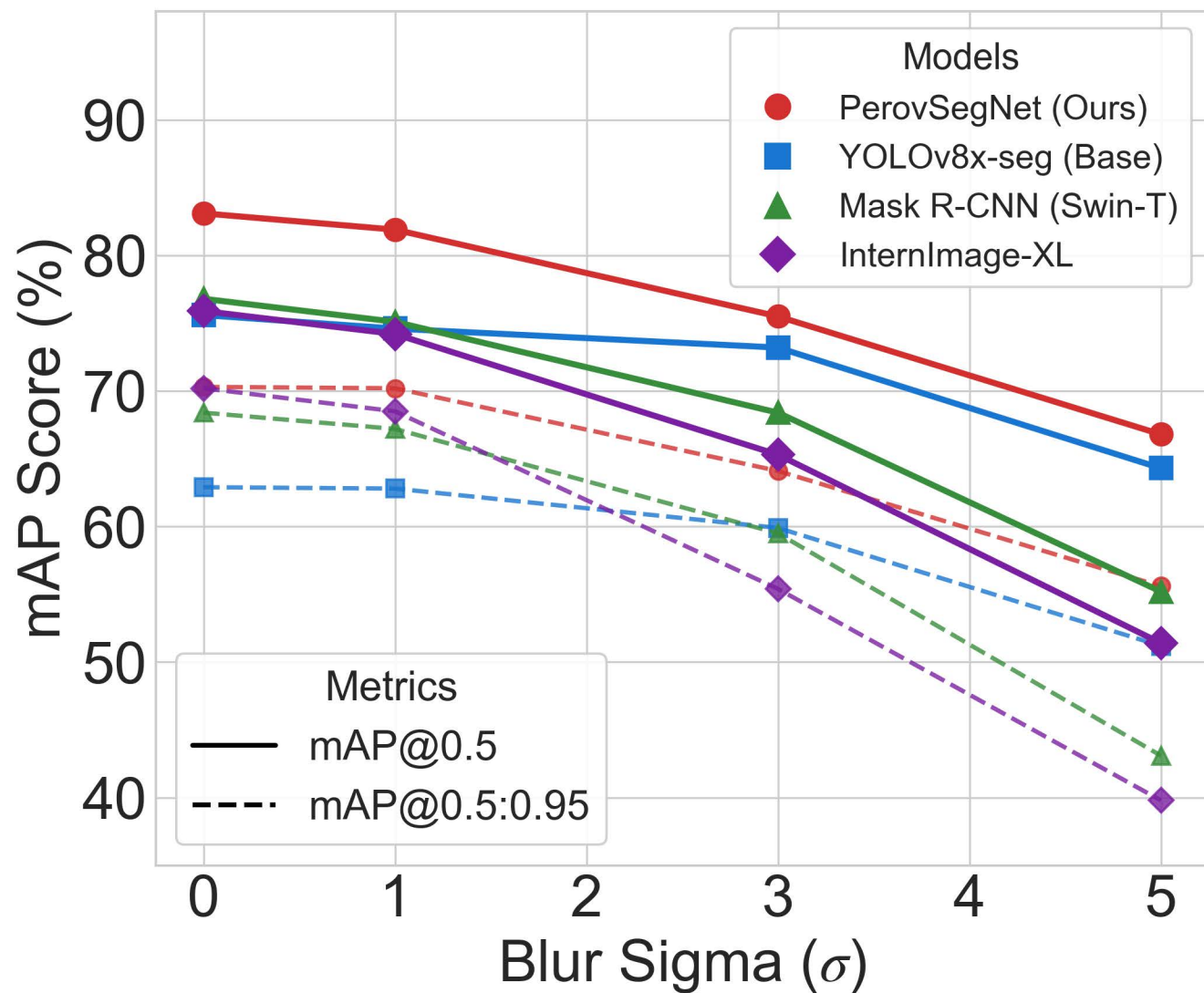
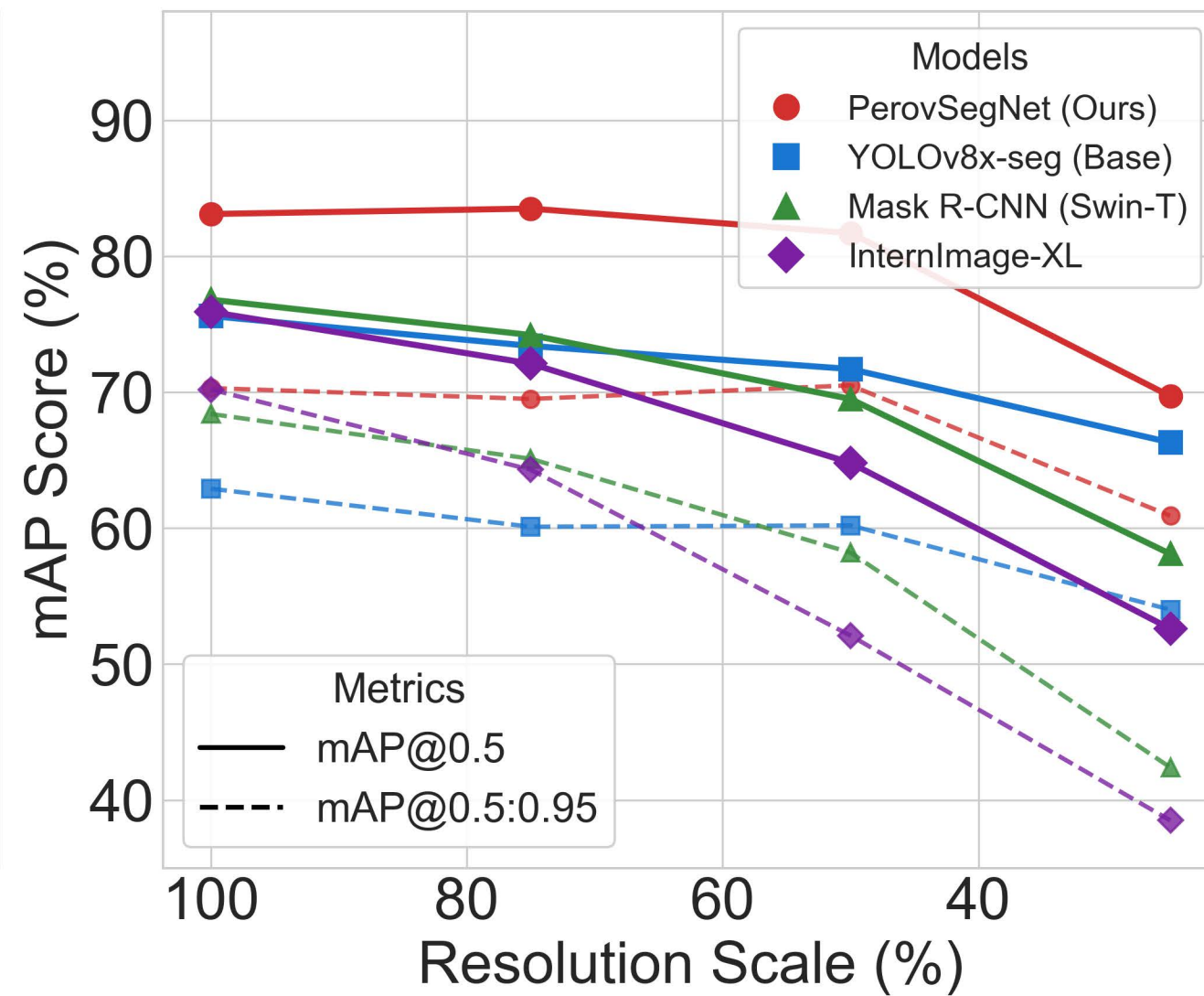
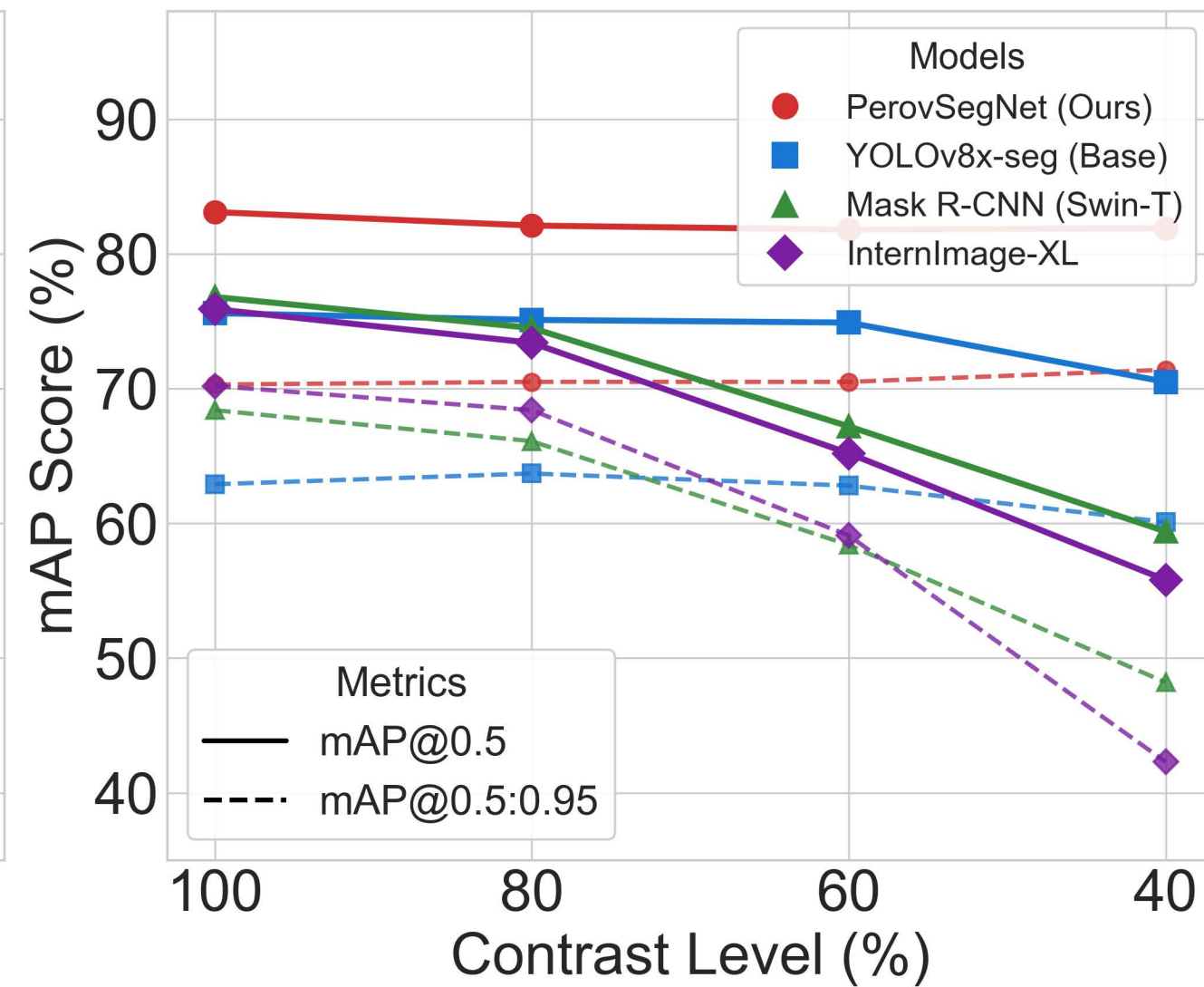


d

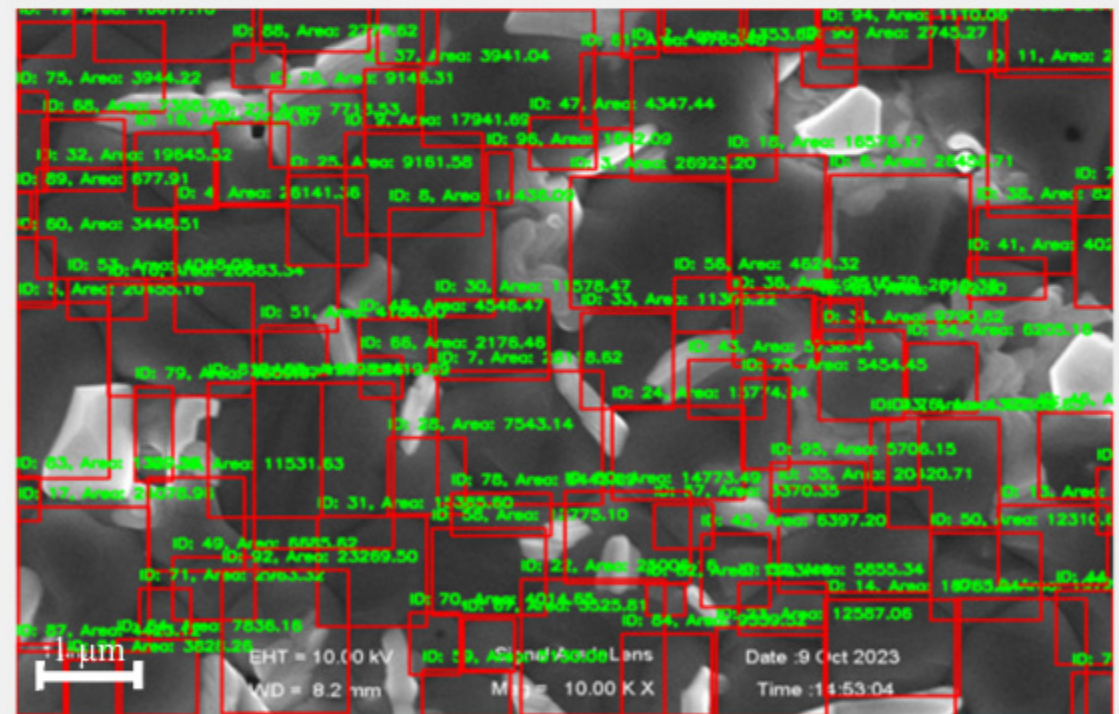
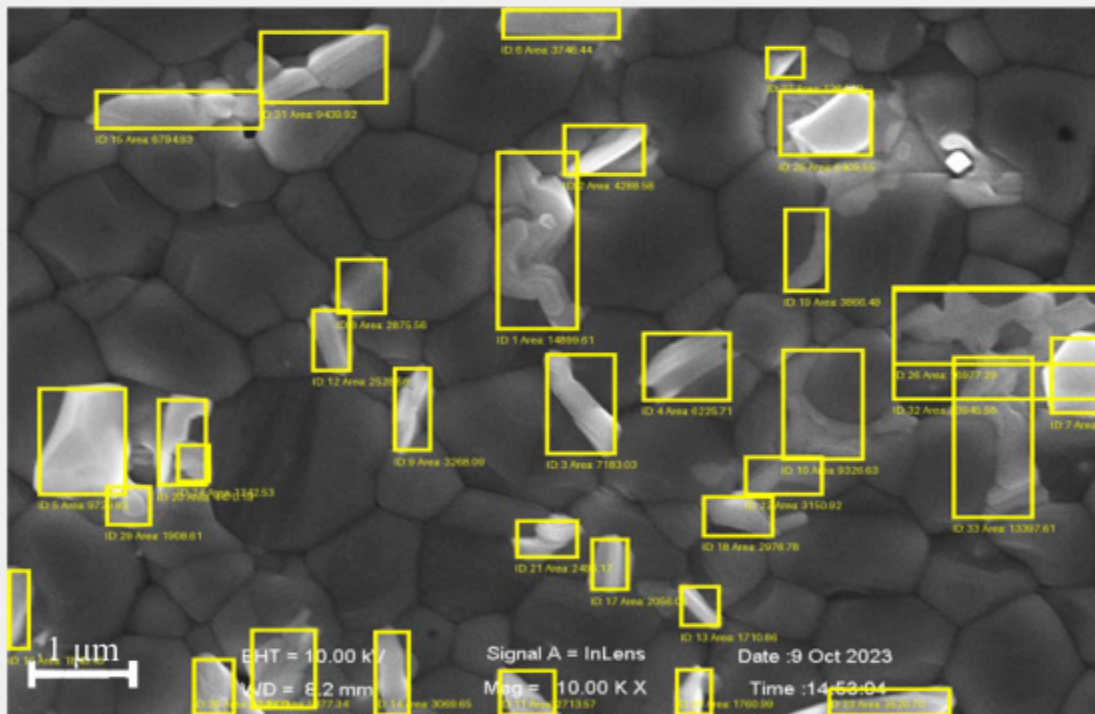
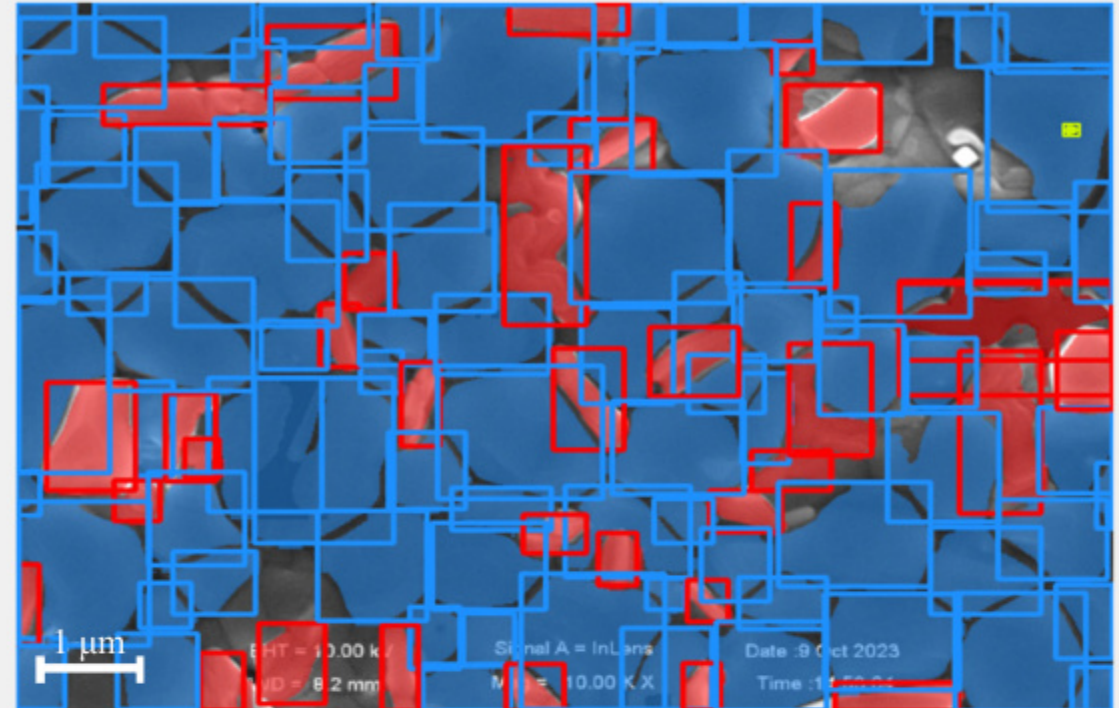
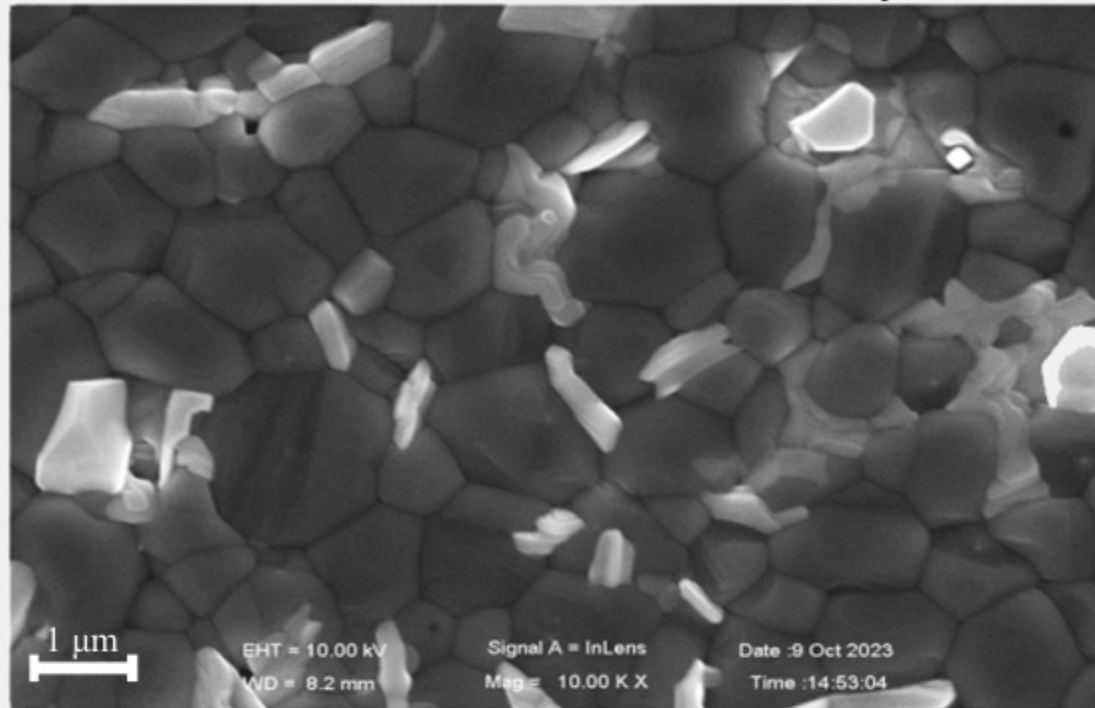


e



(a) Blurring Resilience**(b) Resolution Stability****(c) Contrast Robustness**

Perovskite And Lead Iodide Automatic System



PerovSegNet

load model

Upload File

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"y": 672.0909423828125,
"width": 288.6689758300781,
"height": 759.2012939453125
}

```

2025-09-12 21:08:07 - Image detection completed! : D:/ultralytics-main/000008.jpg

Confidence threshold

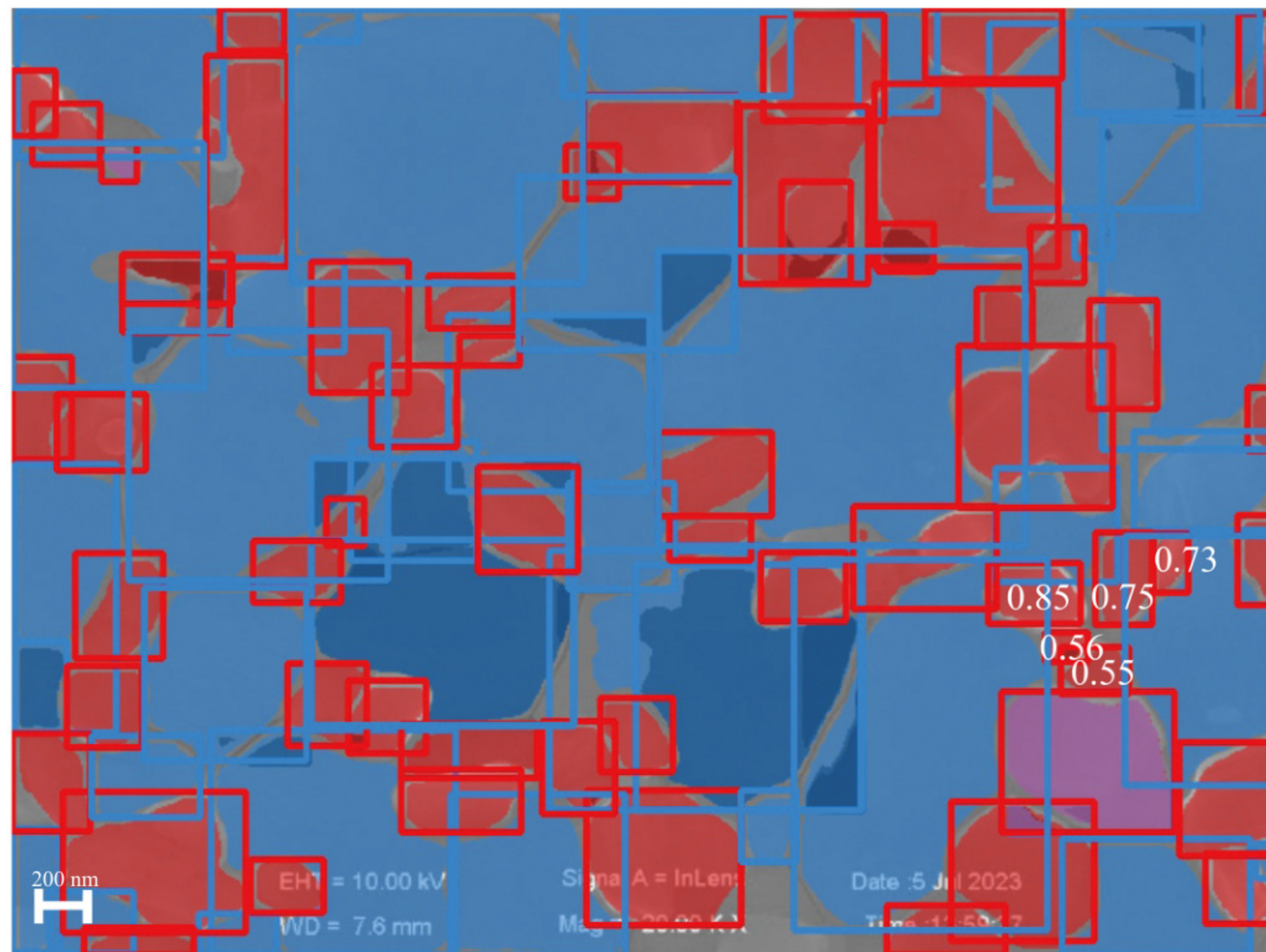


0.50

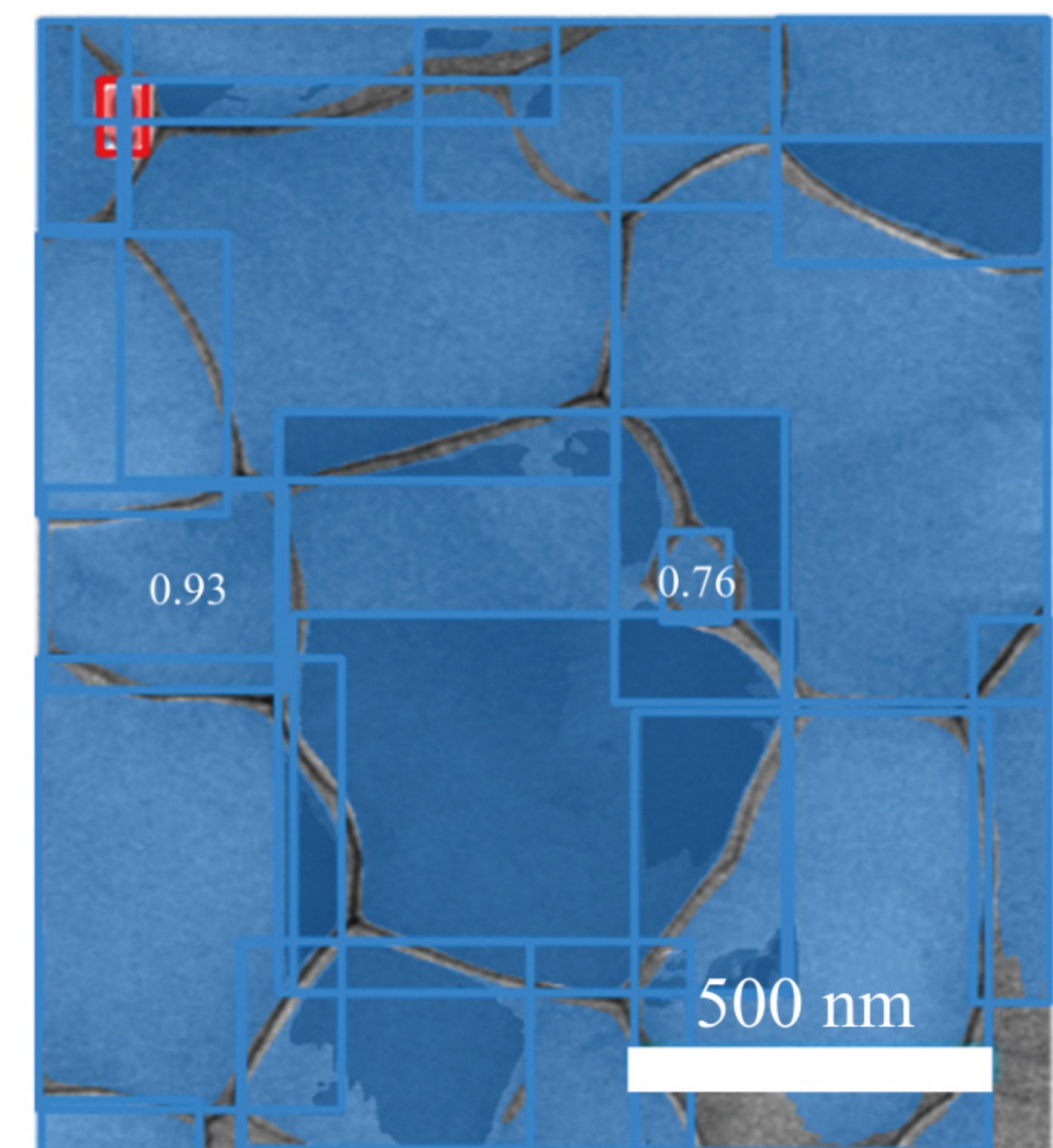
Start

Stop

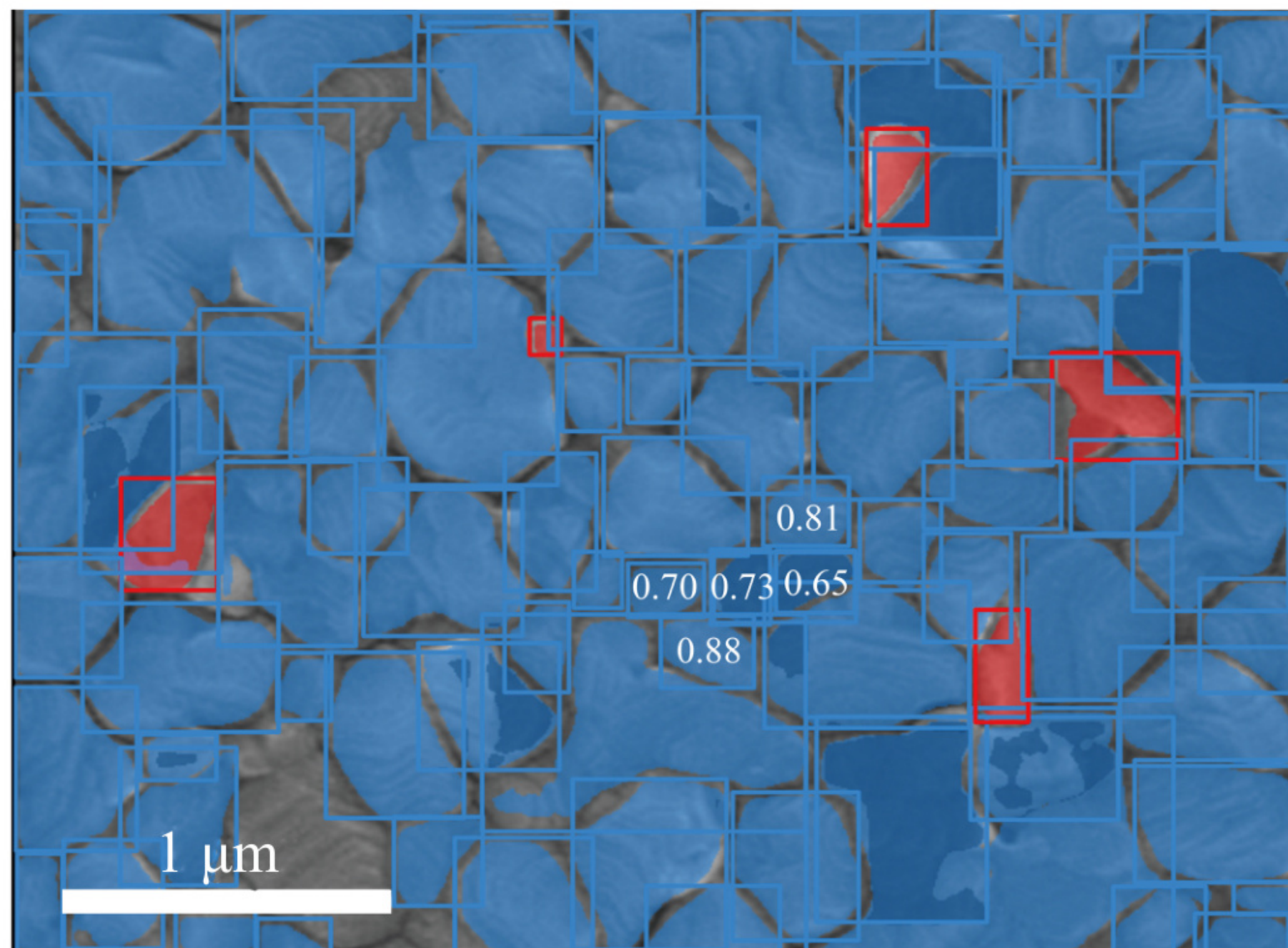
a



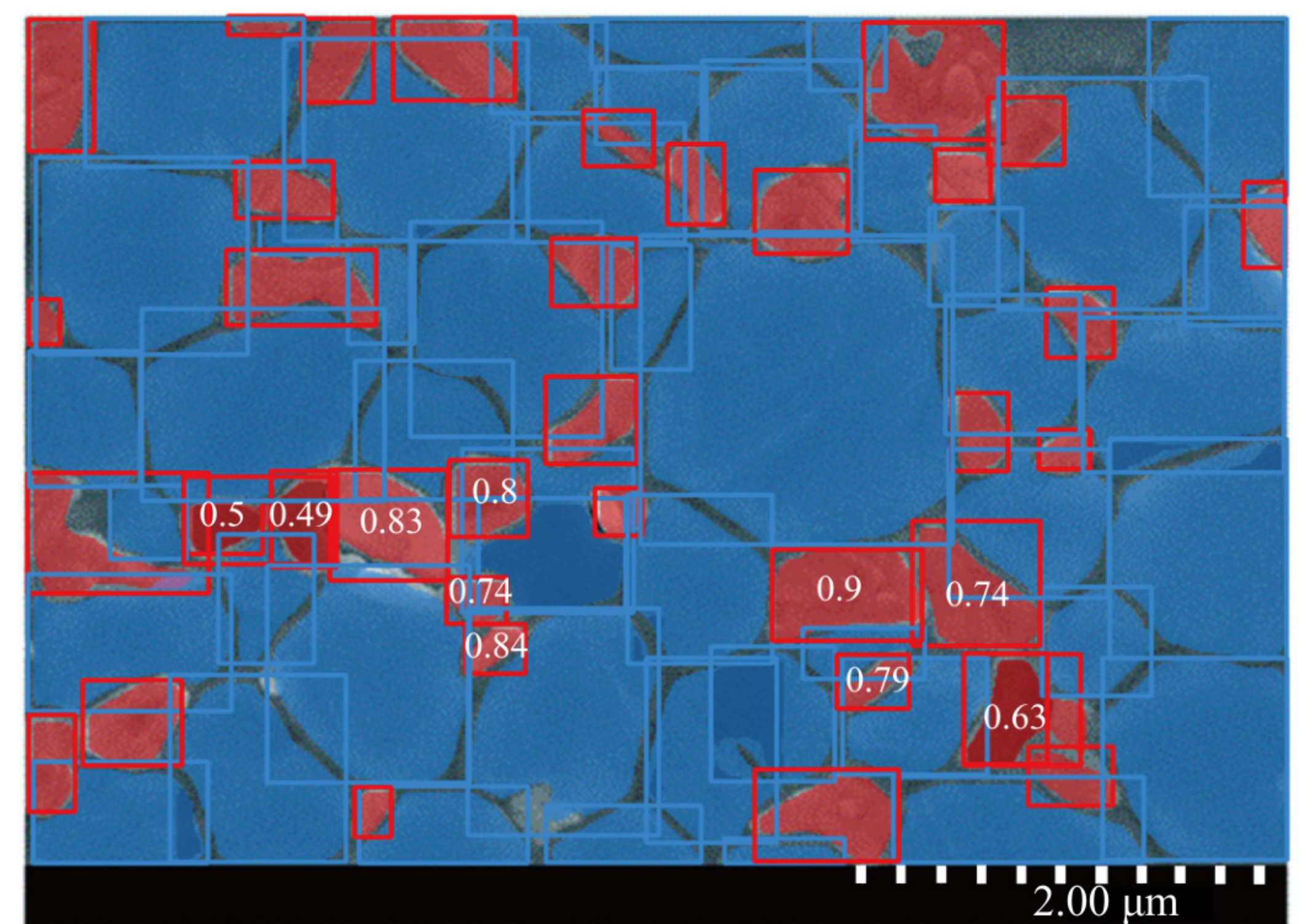
b

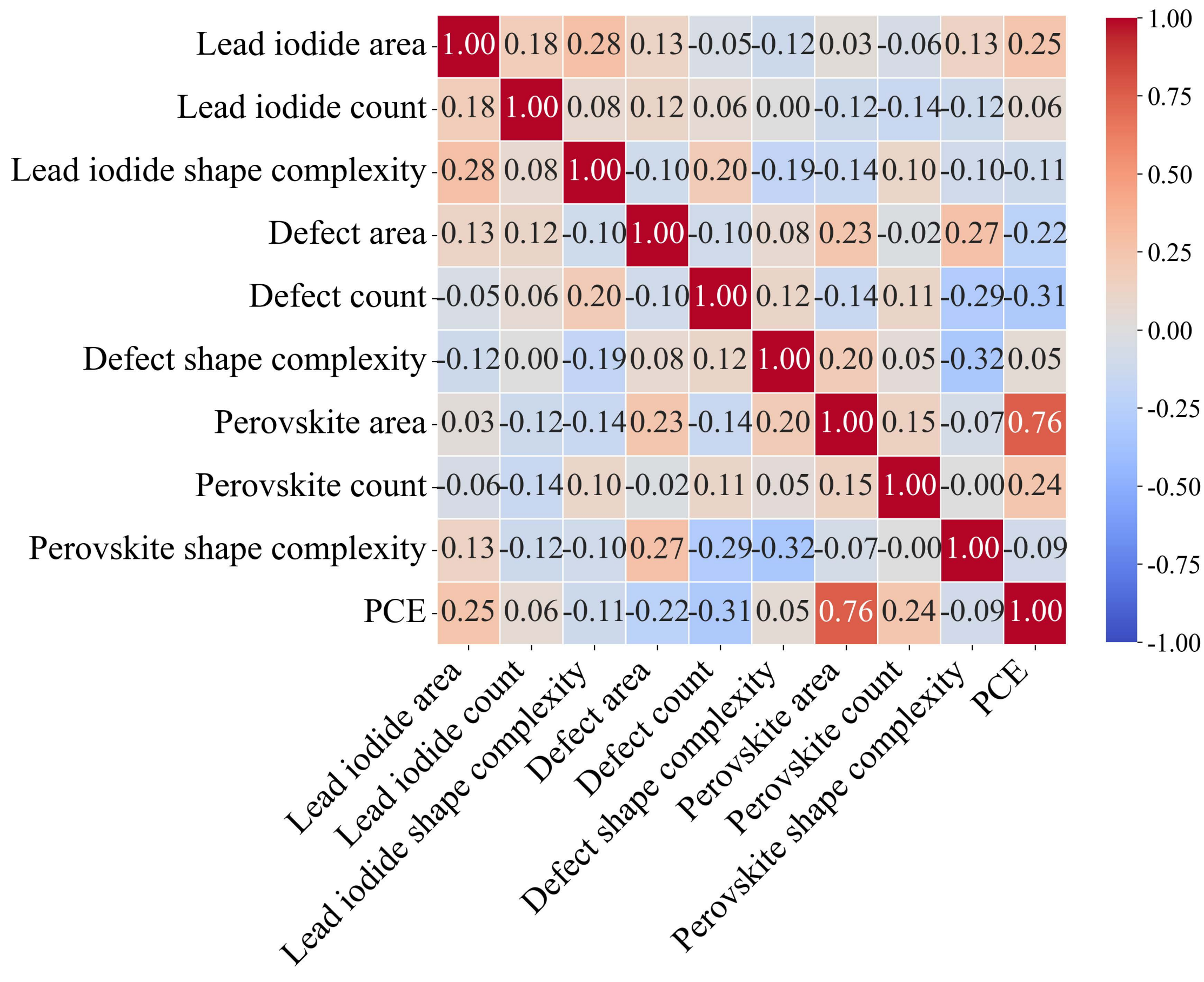


c

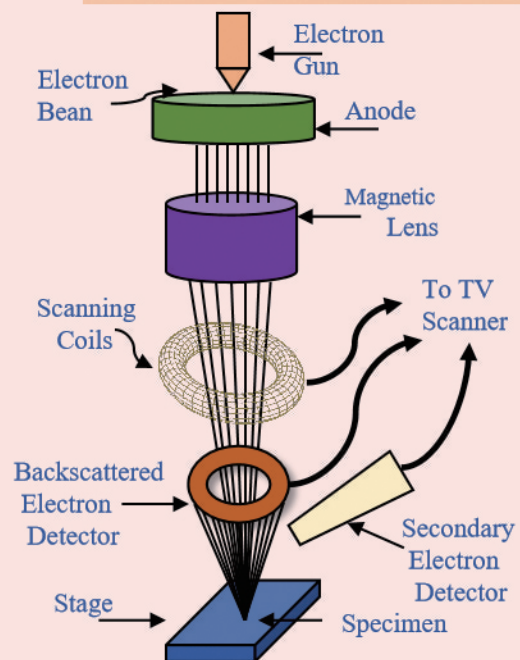


d

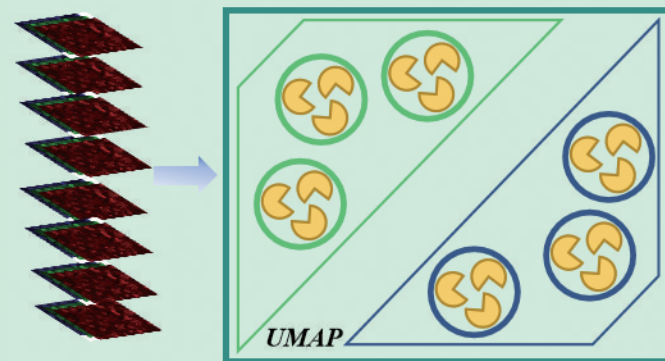




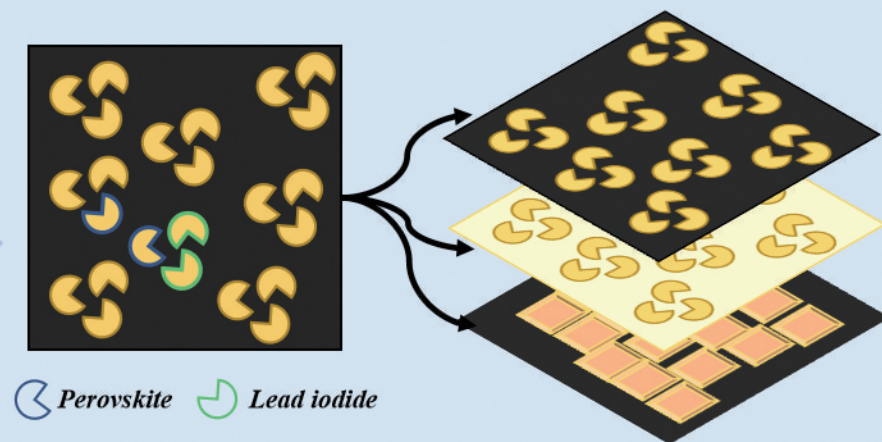
1. SEM image acquisition



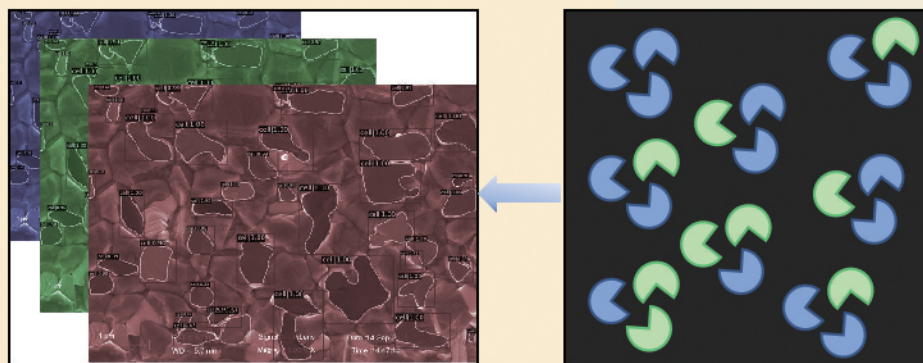
2. Molecule assessment



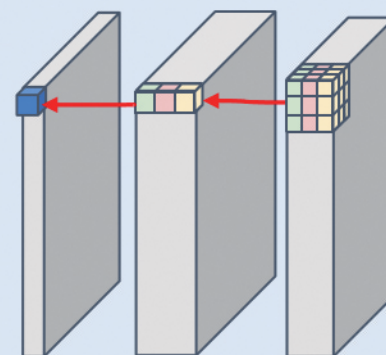
3. Annotation & Augmentation



5. Image interference & Object segmentation

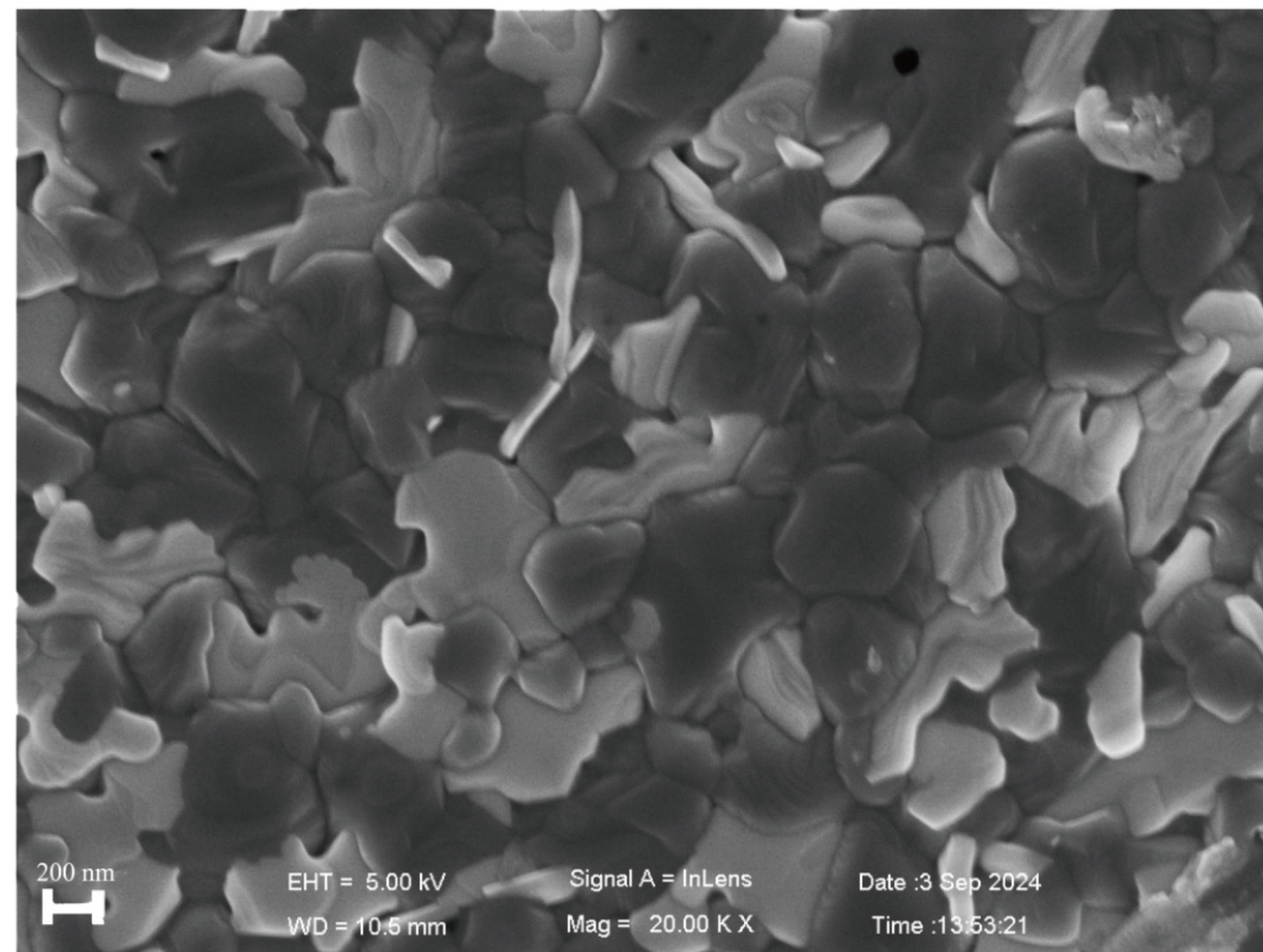


4. PerovSegNet model training

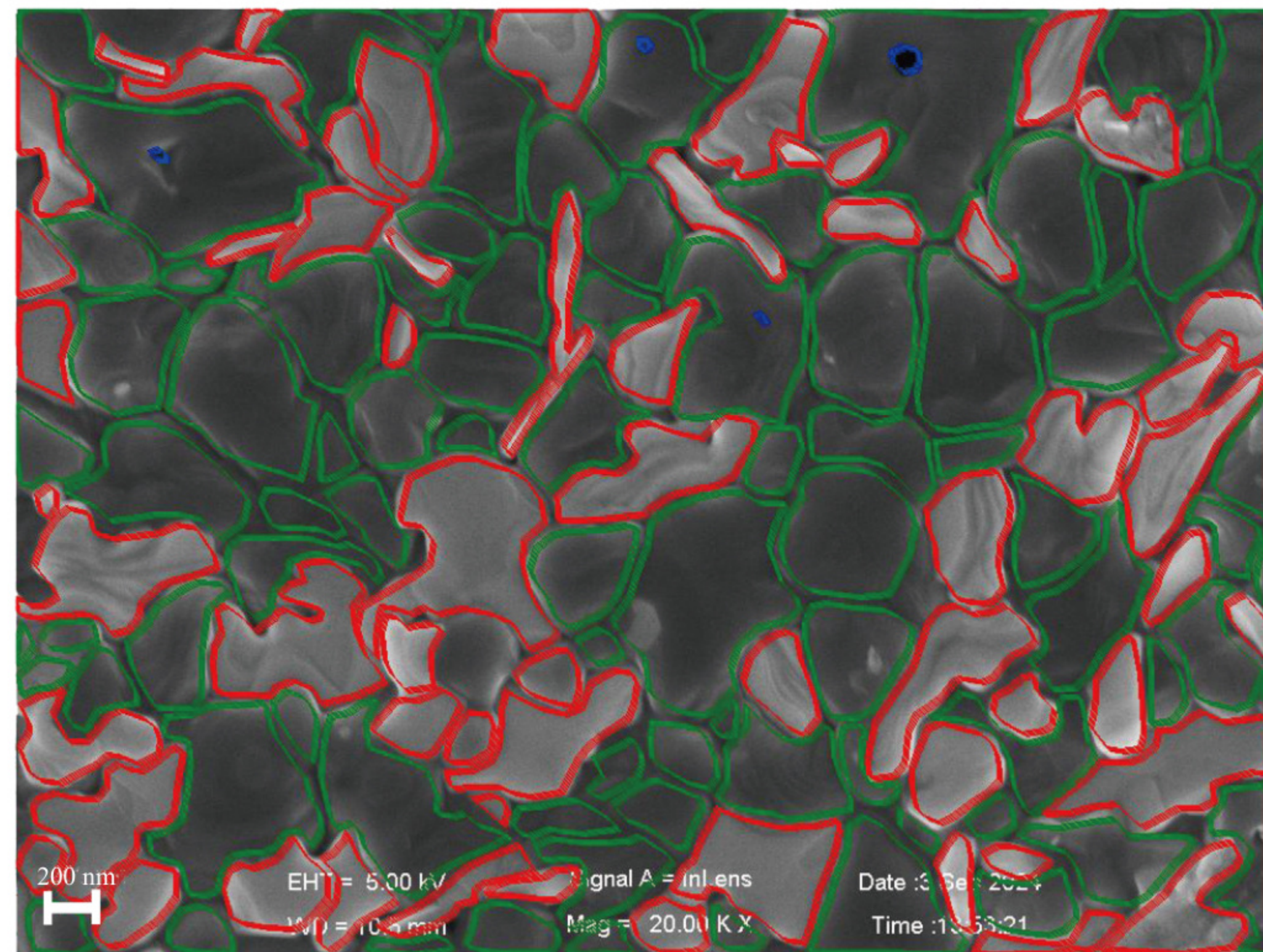


Dataset

a



b



c

